Theoretical Analysis of Blockchain-based Proportional Fair Scheduling for O-RAN

***Abstract*—Blockchain-based proportional fair scheduling (BC-PFS) provides a decentralized and trustworthy platform for cross-operator resource sharing and scheduling in O-RAN. In the BC-PFS, blockchain consensus process as an integral part of scheduling process may induce scheduling delay and consensus failure, both of which are nonnegligible factors in BC-PFS for O-RAN. Yet their impacts on the performance of BC-PFS in O-RAN remain unexplored. This paper establishes an analytical framework for BC-PFS and investigates the impact of scheduling delay and consensus failure on system performance. We first develop a mathematical model of BC-PFS and prove the weak convergence of the scheduling process considering the scheduling delay caused by relatively long block time. Moreover, we derive the semi-closed-form expression of user average throughput, along with its approximation and lower bound. We extend the model to evaluate throughput performance with both scheduling delay and consensus failure. Furthermore, we reveal the trade-off between these two factors and derive the optimal scheduling interval. We also investigate the joint influence of channel coherence time and blockchain propagation delay on the optimal scheduling interval. The pooling effect is quantified by comparing the network utility of BC-PFS in O-RAN with that of conventional non-cooperative scheduling schemes. Finally, we conduct simulations to validate our proposed model and theoretical findings.**

***Index Terms*—Blockchain, O-RAN, proportional fairness, trustworthy, scheduling.**

I. Introduction

Open radio access network (O-RAN) emerges as a transformative paradigm characterized by openness, programmability, intelligence, and disaggregated architecture with interoperable interfaces [1], [2]. A major objective of O-RAN is to facilitate wireless resource sharing among multiple participants [3]–[5], which increases the demands on efficient and trustworthy coordination across networks or operators in O-RAN [6], [7]. However, the trust foundation for cross-operator collaborative scheduling is often limited in O-RAN due to its openness and multi-stakeholder nature [8], [9]. In particular, without trusted mechanisms for coordinating scheduling decisions and ensuring reliable execution across networks, operators may have concerns about potential deceptive behaviors and become reluctant to share network resources and services [10]. These challenges highlight the pressing need for trustworthy scheduling paradigms that enable reliable collaboration in multi-operator O-RAN.

To fill this gap, blockchain, a critical enabler for establishing trust among multiple parties through immutable records and verifiable smart contract execution [11], [12], gives rise to the blockchain-based proportional fair scheduling (BC-PFS) for trustworthy cross-domain scheduling in O-RAN. The BC-PFS provides an explicit trust mechanism required to enable reliable collaboration across heterogeneous networks in O-RAN. Specifically, the BC-PFS records authenticated user reports, scheduling process, and allocation results on a shared, distributed, and immutable ledger, thus offering multiple operators a consistent and traceable view of coordination in O-RAN [13]. Smart contracts further enable automatic and verifiable execution of scheduling strategies coded on-chain [14]. Therefore, recent studies [15]–[17] have explored blockchain-enhanced inter-network collaboration to facilitate cross-operator resource coordination. Our previous work [xx] has implemented BC-PFS through smart contracts, and demonstrated its efficiency and utility gains in O-RAN through blockchain-enabled cross-operator scheduling.

The BC-PFS enables efficient and trustworthy cross-operator scheduling in O-RAN through blockchain. However, the modeling of such blockchain-enabled scheduling across networks has been underdeveloped. As a result, the network behaviors ~~and performance remain unclear, and impacts of most key factors on BC-PFS in O-RAN are unexplored~~. Specifically, the scheduling process of BC-PFS inherently involves blockchain consensus process, whose nonnegligible impact on BC-PFS performance has been ignored [18], [19]. In the BC-PFS, the coordinated scheduling decisions generated by blockchain are delivered to the O-RAN distributed units (O-DUs) for radio resource allocation. To ensure that scheduling decisions are made based on trusted historical throughput, the scheduling results and updated throughput in each scheduling round should be validated through consensus before next round starts. Accordingly, the scheduling interval is determined by the block time, i.e., the average time to create a new block that contains the latest scheduling result. Since each scheduling decision remains fixed over a scheduling interval, a longer block time increases the likelihood that the channel state changes during the scheduling interval [20]. Therefore, if the block time exceeds the channel coherence time, the scheduling decision may become outdated as the channel conditions vary in O-RAN. In contrast, an excessively short block time increases the risk of consensus failure [21]. Both cases can degrade scheduling performance, which calls for a proper block time for BC-PFS in O-RAN to trade off these impacts.

On the one hand, the configured scheduling interval, or equivalently the block time, may be longer than the channel coherence time. As a result, the BC-PFS in O-RAN has to rely on outdated channel state information, which may no longer reflect the latest network conditions. Therefore, blockchains with short block time are more suitable for BC-PFS in O-RAN. For instance, Kaspa achieves a block time of 100 ms [22], while MegaEth and Yona reduce the block time to the order of 10 ms [23], [24]. Projects such as Magicblock and Final DeFi can generate blocks every 1 ms [25], [26], which approaches real-time operations. Although several blockchain platforms claim millisecond-level block times, reliably achieving such

short block time in practice remains challenging. Therefore, the block time may still be longer than the channel coherence time [27], [28]. This mismatch becomes more pronounced in high-mobility scenarios, where rapid fluctuations in channel state amplify the negative impact of delayed scheduling on the BC-PFS performance. On the other hand, excessively shortening the scheduling interval to mitigate scheduling delay may leave insufficient time for consensus and thus lead to consensus failure for BC-PFS in O-RAN. Specifically, the actual block propagation delay is variable, and blockchain maintainers may fail to complete the current round of consensus within the expected block time [29]. Such a failure prevents the current scheduling round from being completed, and results in invalid throughput for that round, degrading overall network performance.

However, it lacks an effective analytical model to capture the scheduling process of BC-PFS. As a result, the performance of BC-PFS in Open RAN remains insufficiently understood, and the impact of the blockchain consensus process on it can not be quantitatively characterized. Meanwhile, how to select an appropriate block time to balance the scheduling delay and consensus failure for BC-PFS deployment in O-RAN is unexplored. Furthermore, blockchain plays a dual role in BC-PFS. The positive impact of the consensus process may offset gains from blockchain-enhanced trustworthy resource pooling. Yet the conditions under which BC-PFS can achieve a positive pooling gain remain unclear.

~~To assess the performance of BC-PFS in O-RAN~~, we ~~develop~~ an analytical model for BC-PFS by considering the impact of consensus process. ~~Moreover, we characterize the user average throughput performance of BC-PFS under scheduling delay and consensus failure.~~ We reveal the trade-off between these two effects and examine their combined impact on network performance. ~~Meanwhile~~, we provide theoretical guidance for selecting an appropriate scheduling interval ~~in the~~ BC-PFS. To evaluate the pooling effect, we quantify the network utility of BC-PFS and compare it with traditional scheduling schemes. Overall, ~~these results~~ provide a theoretical basis for ~~enhancing the practical performance~~ of BC-PFS in O-RAN. The main contributions of this paper are summarized as follows:

- We establish a mathematical model ~~for analyzing BC-PFS in O-RAN and capture the scheduling process across multiple operators~~. ~~Based on this model, we provide a basis to evaluate and enhance the performance of BC-PFS considering~~ the impact of blockchain consensus process.
- We ~~first model~~ the impact of scheduling delay on BC-PFS ~~in O-RAN~~. We prove the weak convergence of the scheduling process ~~based on~~ stochastic approximation ~~theory. We derive the semi-closed-form expression for~~ the ~~user average~~ throughput, along with its approximation and ~~the theoretical bound~~ when the scheduling interval tends to infinity.
- We further analyze the impact of blockchain consensus failure and reveal the trade-off between ~~scheduling delay and consensus failure~~. We ~~derive~~ the optimal scheduling interval that balances their impact and investigate the variation of the optimal scheduling interval with respect to coherence time and blockchain propagation delay.
- ~~Furthermore, we~~ examine the pooling effect introduced by BC-PFS in O-RAN ~~through network utility analysis~~. We derive necessary and sufficient conditions for achieving a positive pooling gain ~~under~~ scheduling delay and consensus failure.
- We validate our model under different network configurations through simulations. The results demonstrate the performance advantages of BC-PFS in O-RAN over non-cooperative scheduling despite the impact of blockchain consensus.

The rest of the paper is organized as follows. Section II establishes the O-RAN model. Section III presents the BC-PFS mechanism. Section IV analyzes the impact of scheduling delay on BC-PFS performance. Section V considers the impact of consensus failure. Section VI identifies the optimal scheduling interval and investigates the trade-off between the scheduling delay and consensus failure. Section VII evaluates the pooling gain. Section VIII provides simulation results, and Section IX concludes the paper.

II. O-RAN Model

A. Multi-operator O-RAN

We consider a multi-operator O-RAN comprising K heterogeneous networks. Each operator network serves N users, so the total number of users is KN, indexed by i $(i = 1, ..., KN)$. The architecture of O-RAN is illustrated in Fig. 1. O-RAN disaggregates base station functions into three components, namely O-RAN central unit (O-CU), O-RAN distributed unit (O-DU), and O-RAN radio unit (O-RU), which can be developed by different vendors or operators. Specifically, the O-CU performs higher-layer RAN functions, while the O-DU handles lower-layer functions, including radio resource scheduling. The O-DU connects to the O-RU, which supports radio transmission and reception. Moreover, O-RAN provides the near-real-time RAN intelligent controller (Near-RT RIC) and non-real-time RIC (Non-RT RIC) to enable programmable coordination and longer-term policy management, respectively.

To enable trustworthy coordination among multiple operators, the BC-PFS introduces a two-layer blockchain architecture, where edge chains support local cross-operator user scheduling and the global chain facilitates coordination across entire network. The detailed architecture of BC-PFS in O-RAN is given in Section III.

Multiple operators leverage open and programmable RAN infrastructures and share a common frequency band through synchronized time-division multiplexing. The instantaneous rate of user i at time t, denoted as $r_i(t)$, is determined by the channel state of its serving operator at that time. We assume that the instantaneous rates $r_i(t)$ and $r_j(t)$ for any two distinct users $i \neq j$ are mutually independent.

B. User Scheduling in O-RAN

In O-RAN, users access the network through the O-RU and report their scheduling-related information. Based on the channel conditions and transmission demands, the O-DU performs

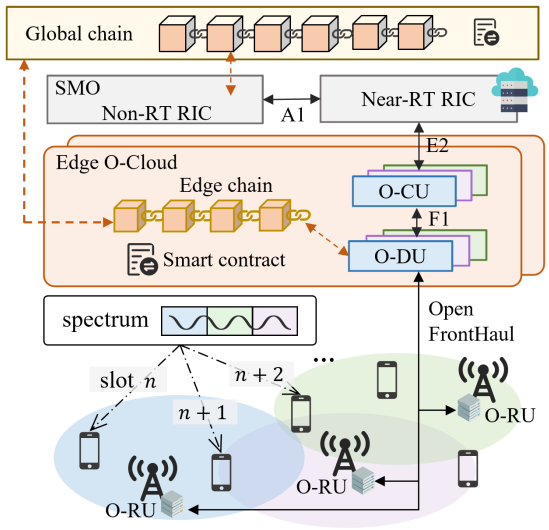


Figure 1. BC-PFS over multi-operator O-RAN.

user scheduling and allocates the radio resources through control signaling. User scheduling follows a time-division framework, where time is divided into uniformly spaced slots. In each slot, the shared spectrum resource is allocated to only one operator, which serves one scheduled user over the bandwidth. This eliminates inter-operator interference and allows the scheduled user's transmission to occupy the entire bandwidth in that slot.

Each slot is defined as one scheduling interval of duration τ. The scheduling decision is made at time $t_n = n\tau$ and applied over the interval $[t_n, t_{n+1})$, which is referred to as slot n. User access to the frequency band is characterized by a binary indicator $I_i(t_n) \in \{0,1\}$, where $I_i(t_n) = 1$ indicates that user i receives transmission service during slot n. Consequently, the achievable rate for user i during slot n is given by $r_i(t) I_i(t_n)$. Furthermore, the throughput achieved by user i up to time t_n is denoted as $\mu_i(t_n)$. We define the long-term average throughput of user i, denoted by $\bar{\mu}_i$, as the limit of the expected instantaneous throughput over time, namely $\bar{\mu}_i = \lim_{n\to\infty} E[\mu_i(t_n)]$.

C. Wireless Channel Model

For user i, its instantaneous signal-to-noise ratio (SNR), denoted as $\mathrm{SNR}_i(t)$, can be modeled as an exponentially distributed random variable. The statistical characteristics of $\mathrm{SNR}_i(t)$ reflect the stochastic nature of channel quality fluctuations. Based on Shannon's theorem, the instantaneous transmission rate $r_i(t)$ of the user exhibits a logarithmic relationship with $\mathrm{SNR}_i(t)$. Let γ_i denote the mean SNR of the user i. Through statistical analysis, the mean $\bar{r}_i$ and variance σ_i^2 of the user transmission rate can be further derived as:

$$\bar{r}_i = \int_0^\infty \log_2(1+\gamma_i x)\exp(-x)\,dx, \tag{1}$$

$$\sigma_i^2 = \int_0^\infty (\log_2(1+\gamma_i x))^2 \exp(-x)\,dx - \bar{r}_i^2, \tag{2}$$

respectively. Here, $\bar{r}_i$ characterizes the long-term stability of the user transmission rate under time-varying channel conditions, while σ_i^2 captures its short-term fluctuations.

We define the channel coherence time T_c, during which the channel state remains stable. Beyond T_c, the channel parameters undergo significant variations. This provides a theoretical foundation for analyzing the coupling relationship between scheduling delay and channel state, particularly in scenarios where blockchain scheduling intervals mismatch with channel coherence time.

III. Blockchain-based Proportional Fair Scheduling

In this section, we present the BC-PFS for O-RAN to enable efficient and trustworthy user scheduling across networks. BC-PFS executes the PFS rule through smart contracts and records scheduling states and allocation results on-chain, which enables verifiable scheduling execution and traceable resource allocation across operators. These capabilities enable the BC-PFS to support trustworthy collaborative user scheduling among multiple stakeholders in O-RAN.

As shown in Fig. 1, the BC-PFS in multi-operator O-RAN introduces a two-layer blockchain architecture. The edge chain, deployed at the edge O-RAN Cloud (O-Cloud), collects periodically reported user channel state information (CSI), executes scheduling algorithm via smart contract, and delivers scheduling results to the O-DU to coordinate resource allocation. It also immutably records scheduling states and allocation results, enabling the scheduling process to be traced and verified. Meanwhile, the global chain aggregates information from edge chains and supports overall coordination, cross-operator settlement, and policy optimization.

In principle, BC-PFS in O-RAN enables trustworthy collaborative scheduling across multiple operators. At each scheduling round, only one user is scheduled per time slot according to the proportional fairness rule. Mathematically, at time t_n, user selection follows the proportional fair rule:

$$i^*(t_n) = \underset{1\le i\le KN}{\operatorname{argmax}} \frac{r_i(t_n)}{\mu_i(t_n)}. \tag{3}$$

Accordingly, the user throughput is updated iteratively to reflect the latest resource allocations:

$$\mu_i(t_{n+1}) = (1-\alpha)\,\mu_i(t_n) + \alpha r_i(t_n) I_i(t_n), \tag{4}$$

where $\alpha \in (0,1)$ is the smoothing factor that determines the weights of historical throughput and newly achieved rate. Smaller α yields gradual throughput adjustments, while larger α enables faster responses to channel fluctuations. Under the proportional fair and throughput update rule, the long-term average throughput converges to the globally optimal solution of the network utility maximization problem [30]:

$$\text{maximize } U \triangleq \sum_{i=1}^{KN} \ln(\bar{\mu}_i). \tag{5}$$

The workflow of BC-PFS consists of the following four stages. 1) The workflow begins with registration, where users and operators are authenticated through the registration contract,

Table I
SYMBOL DEFINITIONS.

Symbols	Explanations
K	Number of operator networks
N	Number of users per operator network
r_i	User rate
μ_i	User throughput
T_c	Channel coherence time
τ	Scheduling interval
T_p	Average blockchain propagation delay
U	Network utility

with their verified identities recorded on-chain. 2) After registration, users periodically report their status, including key metrics like CSI and bandwidth requirements, to the O-DU. The O-DU then forwards the corresponding user states to the edge chain. The status reporting contract aggregates and records user rates for subsequent scheduling. 3) During each scheduling interval, the scheduling contract retrieves the latest user rates and historical throughput records. It then executes the PFS rule to determine the scheduled user and update the throughput records. The scheduling results are recorded in the new block and then delivered to the corresponding O-DU, which performs the radio resource allocation. 4) Finally, the edge chain aggregates scheduling service records and periodically synchronizes them with the global chain. The settlement contract calculates costs based on operator pricing rules and performs settlement among participants. The global information can be further used to support long-term policy optimization and parameter adjustment by the RIC.

Note that each scheduling round should be confirmed on-chain before the next round begins. This ensures that every scheduling decision is based on the trustworthy throughput state jointly validated by operators in O-RAN. However, the scheduling interval, determined by the block time, often exceeds the channel coherence time. The scheduling decision made at the beginning of the scheduling interval remains fixed during the interval despite channel variations, resulting in scheduling delay and suboptimal decisions. Therefore, it is necessary to establish an analytical model that captures this mismatch and quantifies its impact on BC-PFS performance in O-RAN.

IV. DELAYED SCHEDULING MODEL

This section models BC-PFS under scheduling delay and conducts the weak convergence analysis of user throughput update process. We derive the semi-closed-form expression for user average throughput, together with its approximations and bounds under different limiting conditions. We have summarized important symbols in Table I.

A. Delayed Scheduling

To ensure that the scheduling decisions accurately capture the instantaneous channel state and achieve the optimal solution specified in (5), the scheduling interval τ is typically set to be smaller than T_c. However, in the BC-PFS, the scheduling interval τ may exceed T_c. This mismatch causes scheduling decisions to become outdated, which compromises the proportional fairness objective and causes degradation in network utility performance. When $\tau \leq T_c$, the BC-PFS is the ideal scheduling scenario where each scheduling decision aligns with channel variations. However, when $\tau > T_c$, the channel undergoes variations within a single scheduling interval. This misalignment prevents the scheduling decisions from tracking channel dynamics in real-time, causing throughput updates based on suboptimal strategies that ultimately degrade long-term average performance. In this case, we should depict this degradation under practical scheduling conditions.

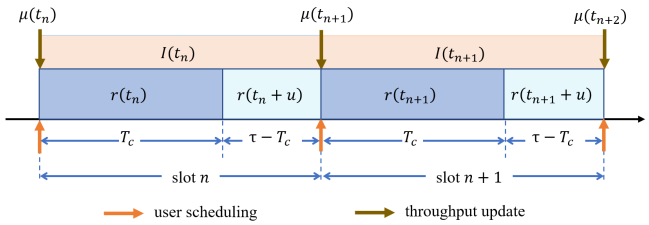


Figure 2. Illustration of BC-PFS delayed scheduling process.

As illustrated in Fig. 2, within a single scheduling interval τ, the user rate r is assumed to be constant during T_c after the scheduling decision and fluctuates thereafter due to channel variations. The scheduling decision is made at the beginning of each scheduling interval, during time slot n, the scheduling decision remains fixed as $I(t_n)$. The throughput is updated at the end of interval, denoted as $\mu(t_{n+1})$. With scheduling delay, throughput updates must aggregate the channel state information within the entire scheduling interval.

From a mathematical modeling perspective, the rule for user selection remains the same as (3). However, the throughput update rule for user i can be extended as:

$$\mu_i^{\alpha}\left(t_{n+1}\right) = \tag{6}$$
$$(1-\alpha)\,\mu_i^{\alpha}\left(t_n\right) + \alpha\frac{1}{\tau}I_i\left(t_n\right)\left(T_c r_i\left(t_n\right) + \int_{T_c}^{\tau} r_i\left(t_n+u\right)du\right),$$

where μ_i^{α} is the sequence related to throughput update with smoothing factor α. The term $T_c r_i\left(t_n\right)$ captures the immediate effect of the current channel condition, while the term $\int_{T_c}^{\tau} r_i\left(t_n+u\right)du$ characterizes the impact of channel variations. The weight $\frac{1}{\tau}$ reflects the weighted average of the user rates in a scheduling interval.

B. Weak Convergence Analysis

To analyze the achievable average throughput in (6) and the impact of scheduling delay, we will first conduct weak convergence analysis of (6) based on the stochastic approximation theory. By examining the weak convergence of this process, the statistical properties of these random perturbations can be translated into the average effects of deterministic ordinary differential equations (ODEs), thereby revealing the algorithm's steady-state performance. Specifically, we construct an associated ODE and prove that (6) weakly converges to the limit point of the ODE as $n \to \infty$. This captures the long-term average behavior of throughput and provides

a rigorous mathematical framework for performance analysis. According to [31], we have the following lemma:

Lemma 1. *Under conditions (C1)-(C7), let $\{\theta_i\}$ be a discrete-time sequence indexed by $\varepsilon \to 0$, namely*

$$\theta_i^{\varepsilon}(t_{n+1}) = \theta_i^{\varepsilon}(t_n) + \varepsilon Y_i^{\varepsilon}(t_n). \tag{7}$$

For any nondecreasing integer sequence q_ε, there exists a subsequence such that the time-shifted process $\theta_i^{\varepsilon}(\varepsilon q_\varepsilon + \cdot)$ converges weakly to a continuous-time process $\theta_i(v)$ satisfying the integral equation:

$$\theta_i(v) = \theta_i(0) + \int_0^v \bar{g}_i(\boldsymbol{\theta}(s))\, ds, \tag{8}$$

where $\bar{g}$ denotes the averaged value of the dynamic process. The conditions are listed as follows:

(C1) $\{Y_i^{\varepsilon}(t_n); \varepsilon, t_n\}$ is uniformly integrable.

(C2) Define $\xi^{\varepsilon}(t_n)$ as the history information up to time t_n. There is a measurable function $g_i^{\varepsilon}(\cdot)$ and a random variable $\beta^{\varepsilon}(t_n)$ such that

$$E_t[Y_i^{\varepsilon}(t_n)] = g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n)) + \beta^{\varepsilon}(t_n), \tag{9}$$

where E_t denotes the expectation conditioned on the history information $\xi^{\varepsilon}(t_n)$.

(C3) $g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n))$ is continuous in $\boldsymbol{\theta}$.

(C4) $\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} E_t[\beta^{\varepsilon}(t_n)] = 0$ in the average sense.

(C5) For $\boldsymbol{\theta}$, the sequence $\{g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n)); \varepsilon, t_n\}$ is uniformly integrable.

(C6) For each $\delta > 0$ there is a compact set A_δ such that $\inf_{t,\varepsilon} P(\xi^{\varepsilon}(t_n) \in A_\delta) \geq 1 - \delta$.

(C7) For $\boldsymbol{\theta}$, there is a continuous-time function $\bar{g}_i(\cdot)$ such that the following expression holds in probability:

$$\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} E_t[g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n)) - \bar{g}_i(\boldsymbol{\theta})] \cdot I_{\{\xi^{\varepsilon}(t_n)\in A_\delta\}} = 0, \tag{10}$$

where $I_{\{\xi^{\varepsilon}(t_n)\in A_\delta\}}$ is the indicator function that equals 1 if $\xi^{\varepsilon}(t_n) \in A_\delta$ and 0 otherwise.

We now construct conditions (C1)-(C7) for the stochastic approximation. Define the measurable function $Y_i^{\alpha}(t_n)$ as:

$$Y_i^{\alpha}(t_n) \triangleq \frac{1}{\tau} I_i(t_n)\left(T_c r_i(t_n) + \int_{T_c}^{\tau} r_i(t_n + u)\, du\right) - \mu_i^{\alpha}(t_n). \tag{11}$$

Starting from the throughput update equation (6), we formulate the iterative sequence in the standard stochastic approximation form (7) as:

$$\mu_i^{\alpha}(t_{n+1}) = \mu_i^{\alpha}(t_n) + \alpha Y_i^{\alpha}(t_n), \tag{12}$$

where $\alpha \to 0$. First, we assume that $\{r_i(t_n), t_n < \infty\}$ is bounded. Therefore, there exists a constant M such that $\sup_{\alpha,t} E[|Y_i^{\alpha}(t_n)|] \leq M$. This implies that $\{Y_i^{\alpha}(t_n); \alpha, t_n\}$ is uniformly integrable, satisfying (C1).

Moreover, based on the wireless channel model established in Section II-C, $\{r_i(t_n), t_n < \infty\}$ constitutes a stationary stochastic process with time-invariant joint distributions. For each bounded random variable $r_i(t_n)$, let $\xi_i(t_n) = \{r_i(\ell) : \ell \leq t_n\}$ denote the historical rate information up to time slot n, and we define

$$h_i^{\alpha}(\boldsymbol{\mu}) \triangleq E_t\left[\frac{1}{\tau} I_i(t_n)\left(T_c r_i(t_n) + \int_{T_c}^{\tau} r_i(t_n + u)\, du\right)\right]. \tag{13}$$

Therefore, we have

$$E_t[Y_i^{\alpha}(t_n)] = h_i^{\alpha}(\boldsymbol{\mu}) - E_t[\mu_i^{\alpha}(t_n)]. \tag{14}$$

In this case, (C2) is satisfied with $\beta^{\alpha}(t_n) = 0$. Besides, since $\beta^{\alpha}(t_n) = 0$ for all conditions, (C4) is also met. Based on the Bayesian decomposition, (13) can be written as:

$$h_i^{\alpha}(\boldsymbol{\mu}) = \frac{1}{\tau}\Pr(I_i(t_n) = 1)\left(T_c E_t[r_i(t_n)|I_i(t_n) = 1] + E_t\left[\int_{T_c}^{\tau} r_i(t_n + u)\, du | I_i(t_n) = 1\right]\right). \tag{15}$$

With the scheduling rule in (3), $h_i^{\alpha}(\boldsymbol{\mu})$ can be expressed as

$$h_i^{\alpha}(\boldsymbol{\mu}) = \frac{1}{\tau}\Pr\left(\frac{r_i(t_n)}{\mu_i(t_n)} > \frac{r_j(t_n)}{\mu_j(t_n)}, \forall j \neq i\right) \cdot \left(T_c E_t\left[r_i(t_n) | \frac{r_i(t_n)}{\mu_i(t_n)} > \frac{r_j(t_n)}{\mu_j(t_n)}, \forall j \neq i\right] + E_t\left[\int_{T_c}^{\tau} r_i(t_n + u)\, du\right]\right). \tag{16}$$

The function $h_i^{\alpha}(\boldsymbol{\mu})$ satisfies local Lipschitz continuity in $\boldsymbol{\mu} \in \mathbb{R}_+^{KN}$. Similarly, owing to the boundedness and stationarity of $\{r_i(t_n), t_n < \infty\}$, $h_i^{\alpha}(\boldsymbol{\mu})$ is continuous in $\boldsymbol{\mu}$ and uniformly integrable. These properties guarantee (C3) and (C5). Meanwhile, according to [30], $\xi_i(t_n)$ can be assumed to reside in a compact sequence space, thereby satisfying (C6).

Leveraging the stationarity of $\{r_i(t_n), t < \infty\}$, we further define the steady-state expectation function for user i:

$$\bar{h}_i(\boldsymbol{\mu}) = \Pr\left(\frac{r_i}{\mu_i} > \frac{r_j}{\mu_j}, \forall j \neq i\right) \cdot \left(\frac{T_c}{\tau} E\left[r_i | \frac{r_i}{\mu_i} > \frac{r_j}{\mu_j}, \forall j \neq i\right] + \left(1 - \frac{T_c}{\tau}\right) E[r_i]\right), \tag{17}$$

where μ_i and r_i denote the canonical representations of $\mu_i(t)$ and $r_i(t)$ that exclude time dependencies, respectively. To establish the asymptotic equivalence between $h_i^{\alpha}(\boldsymbol{\mu})$ and $\bar{h}_i(\boldsymbol{\mu})$, we utilize the ergodicity of the stationary process. As t_n is sufficiently large, the dependence of $h_i^{\alpha}(\boldsymbol{\mu})$ on historical data approaches zero. By Khinchine's law of large numbers, for dependent stationary processes that satisfy the asymptotic independence, the average of conditional expectations converges to the steady-state expectation. Therefore, $h_i^{\alpha}(\boldsymbol{\mu})$ converges to the $\bar{h}_i(\boldsymbol{\mu})$ in probability:

$$\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} \left[h_i^{\alpha}(\boldsymbol{\mu}) - \bar{h}_i(\boldsymbol{\mu})\right] = 0. \tag{18}$$

Consequently, (C7) can be assured. At this point, all conditions of Lemma 1 on stochastic approximation are verified. The throughput update equation (12) converges weakly to a continuous-time process. According to (8), we take the time

derivative of the continuous-time throughput update equation, which leads to the following theorem:

Theorem 1. *For the BC-PFS with scheduling delay, as $n \to \infty$, the user throughput $\mu_i(t_n)$ weakly converges to the limit point of the solution to the following ODE system:*

$$\dot{\mu}_i = \bar{h}_i(\boldsymbol{\mu}) - \mu_i, i = 1, \ldots, KN, \tag{19}$$

where $\bar{h}_i(\boldsymbol{\mu})$ is the conditional expectation of user rate r_i under event $\frac{r_i}{\mu_i} > \frac{r_j}{\mu_j}, \forall j \neq i$, with expression given in (17)*. Furthermore, employing ODE asymptotic analysis similar to [30], the limit point of* (19) *is unique and independent of initial conditions, which equals $\bar{\mu}_i$.*

Theorem 1 transforms the stochastic BC-PFS process into a deterministic differential equation for analysis. It demonstrates that $\mu_i(t_n)$ weakly converges to the unique asymptotically stable limit point of ODE $\bar{\mu}_i$ as $n \to \infty$. This provide a crucial mathematical foundation for the performance analysis of BC-PFS.

C. Throughput Analysis

This subsection aims to derive semi-closed form expressions for the long-term average throughput under scheduling delays. Building upon the weak convergence property of (6), we decompose expectation terms and leverage channel statistics to establish explicit relationships between throughput and network parameters.

Theorem 2. *With the Rayleigh fading channel, for user i in the BC-PFS, the approximation of its long-term average throughput is*

$$\tilde{\mu}_i(\tau; T_c) = \frac{T_c}{KN\tau}\left(\bar{r}_i + \sigma_i g(KN, \tfrac{\bar{r}_i}{\sigma_i})\right) + \frac{1-\frac{T_c}{\tau}}{KN}\left(\bar{r}_i \Phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right) + \sigma_i \phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)\right). \tag{20}$$

where $g(N, \eta) \triangleq N \int_{-\eta}^{\infty} x\phi(x)\Phi(x)^{N-1}dx$. The function $\phi(\cdot)$ and $\Phi(\cdot)$ are the PDF and CDF of the standard normal distribution, respectively. $\bar{r}_i$ and σ_i are the mean and standard deviation of r_i, namely (1) *and* (2)*.*

Proof. Please see Appendix A. ∎

When $\tau = T_c$, each throughput update utilizes the most recent scheduling information. This finding is consistent with the conclusion of previous work [32]. As $\frac{\tau}{T_c}$ increases from 1, the weight $\frac{T_c}{\tau}$ associated with the first term gradually weakens, whereas the weight $1-\frac{T_c}{\tau}$ associated with the second term increases. Furthermore, as $\frac{\tau}{T_c} \to \infty$, the contribution of the first term vanishes, while the second term dominates the average throughput. This indicates that the throughput gain obtained from instantaneous scheduling information gradually diminishes as the scheduling delay.

To simplify the user throughput derivation in (20), we now consider the asymptotic case where $\gamma_i \to \infty$. This limiting condition allows us to derive semi-closed form expressions while maintaining key insights about the system's behavior under extreme channel conditions.

Corollary 1. *When $\gamma_i \to \infty$, for user i in the BC-PFS, its average throughput can be approximated by:*

$$\tilde{\mu}_i^{\infty}(\tau; T_c) = \frac{1}{KN}\left(\bar{r}_i + \frac{T_c}{\tau}\sigma_i g(KN, \infty)\right). \tag{21}$$

Proof. Please see Appendix B. ∎

The average throughput with scheduling delay in the BC-PFS comprises two distinct components: the baseline throughput $\frac{\bar{r}_i}{KN}$, which reflects equal resource allocation among users, and multiuser diversity gain from channel variations. Notably, when the scheduling delay factor $\tau = T_c$, the system achieves maximum scheduling gain equivalent to the PFS adopting accurate channel information. As τ increases, performance degradation due to delayed scheduling becomes progressively apparent.

Meanwhile, for the fixed channel coherence time T_c, as the scheduling interval $\tau \to \infty$, nearly all scheduling decisions become suboptimal, resulting in average throughput degradation. In this case, we obtain the following corollary.

Corollary 2. *When $\tau \to \infty$, for user i in the BC-PFS, its average throughput can be approximated by:*

$$\lim_{\tau \to \infty} \tilde{\mu}_i(\tau; T_c) = \frac{1}{KN}\left(\bar{r}_i \Phi\left(\frac{\bar{r}_i}{\sigma_i}\right) + \sigma_i \phi\left(\frac{\bar{r}_i}{\sigma_i}\right)\right). \tag{22}$$

Proof. Please see Appendix C. ∎

As the network scale remains constant and $\frac{\tau}{T_c}$ increases, the impact of timely scheduling on throughput updates progressively diminishes and the proportion of outdated scheduling information used in throughput updates becomes larger, which exacerbates the average throughput degradation. The result (22) equals the limit of (20) as $\tau \to \infty$ and represents the lower bound of average throughput.

V. Consensus Failure

Section IV has analyzed the performance degradation of BC-PFS caused by scheduling delay, which can be mitigated by reducing scheduling interval τ toward T_c. However, when the scheduling interval τ becomes excessively small, the consensus process may fail to complete within a scheduling round, which results in consensus failure and undermines throughput efficiency. This section explores the degradation in the average throughput caused by consensus failures.

A. Probability of Consensus Failure

In the BC-PFS with Raft consensus mechanism, the consensus delay mainly consists of two message exchange phases, including the proposal propagation from the leader to followers and the acknowledgment collection from followers [33], [34]. When the actual consensus delay is longer than the predetermined scheduling interval τ, blockchain nodes may fail to receive critical messages in time. This prevents the current scheduling round from completing successfully, which results in no effective service for users during that round and consequently degrades their average throughput. Since each message propagation delay can be modeled as an exponential random variable [35], we adopt a two-stage Erlang distribution with average propagation delay T_p to characterize consensus

delay. In this case, the probability of timeout event, namely the probability of consensus failure, can be expressed as:

$$q_f = \exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right). \tag{23}$$

Consequently, the probability of successful scheduling is expressed as $1-q_f$. From (23), both the scheduling interval τ and the propagation delay T_p jointly influence the consensus failure probability q_f. Since $\frac{\partial q_f}{\partial \tau} = -\exp\left(-\frac{\tau}{T_p}\right)\frac{\tau}{T_p^2} < 0$, increasing the scheduling interval τ leads to a reduction in the consensus failure probability. A longer scheduling interval provides more time for nodes to achieve consensus and decreases the risk of failure. Additionally, since $\frac{\partial q_f}{\partial T_p} = \exp\left(-\frac{\tau}{T_p}\right)\frac{\tau^2}{T_p^3} > 0$, an increase in propagation delay T_p reduces the speed of information exchange among nodes and diminishes the efficiency of the consensus. This inefficiency increases the probability of consensus timeouts, which corresponds to the consensus failure probability.

B. Impact of Consensus Failure

Integrating the consensus failure model in Section V-A into the BC-PFS framework, we revise the user throughput update in (6) to reflect the loss caused by consensus failure. The updated throughput for user i at time t_{n+1} becomes:

$$\mu_i(t_{n+1}) = (1-\alpha)\,\mu_i(t_n) + \alpha(1-q_f)\frac{1}{\tau} \cdot I_i(t_n)\left(T_c r_i(t_n) + \int_{T_c}^{\tau} r_i(t_n+u)\,du\right). \tag{24}$$

This formulation explicitly couples the impact of blockchain consensus failure with scheduling delay caused by channel variations in the BC-PFS. Since this modified expression (24) introduces only additional terms that are independent of both the transmission rate and throughput when compared to (6), it inherently preserves the weak convergence property while maintaining the uniqueness of the average throughput convergence result. Based on Theorem 2, we derive the semi-closed form average user throughput under both scheduling delay and consensus failure:

$$\tilde{\mu}_i^{\text{fail}}(\tau;T_c) = \left(1-\exp\left(-\tfrac{\tau}{T_p}\right)\left(1+\tfrac{\tau}{T_p}\right)\right)\tilde{\mu}_i(\tau;T_c). \tag{25}$$

Based on Corollary 1, when $\gamma_i \to \infty$, i.e. $\frac{\bar{r}_i}{\sigma_i} \to \infty$, (25) can simplify to

$$\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c) = \left(1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)\right) \tag{26}$$

$$\cdot\,\frac{1}{KN}\left(\bar{r}_i + \frac{T_c}{\tau}\sigma_i g(KN,\infty)\right), \tag{27}$$

which demonstrates that average throughput is reduced due to consensus failure and delayed state information. The term $1-q_f$ reflects the impact of consensus failure. Meanwhile, the term $\frac{T_c}{\tau}$ captures the throughput loss caused by the outdated information. The scheduling interval τ establishes an inherent trade-off that shorter intervals enable more frequent scheduling updates but elevate the probability of consensus failure.

VI. Scheduling Delay versus Consensus Failure

Building upon the analytical model of BC-PFS in Section IV and Section V, this section investigates the optimal scheduling interval τ that balances the trade-off between scheduling delay and consensus failure in the BC-PFS. Analyzing the average throughput (26) as a function of τ, we reveal the combined influence of channel coherence time T_c and blockchain propagation delay T_p on the optimal scheduling interval.

A. Scheduling Interval Optimization

In this subsection, we optimize the scheduling interval τ to balance the consensus failure probability and the scheduling delay. These two factors directly influence the average throughput. Specifically, the optimal scheduling interval is obtained by maximizing the average user throughtput considering both scheduling delay and consensus failure, which can be formulated as

$$\max_{\tau} \quad \tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c). \tag{28}$$

The following theorem specifies the optimal value of the scheduling interval to balance consensus failure and channel variation for maximum throughput.

Theorem 3. *The function $\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)$ has a unique stationary point $\tau^\star$ that satisfies $\partial\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)/\partial\tau\,|_{\tau=\tau^\star}=0$ and achieves its global maximum at this point.*

Proof. Please see Appendix D. ■

Theorem 3 demonstrates that, under general conditions, the average throughput $\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)$ with respect to the scheduling interval τ exhibits a unique stationary point corresponding to its global maximum, which reveals the trade-off between scheduling delays and consensus failure probabilities. Specifically, the average throughput exhibits distinct characteristics in different intervals. Over the range $(0,\tau^\star]$, where blockchain propagation delay dominates the system performance, increasing the scheduling interval τ effectively enhances the probability of successful scheduling and mitigates performance degradation caused by propagation delay T_p, which results in a monotonically increasing trend of $\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)$. Conversely, in the interval $[\tau^\star,+\infty)$, the time-varying nature of wireless channels becomes the predominant factor, where longer scheduling intervals lead to outdated channel state information for scheduling decisions, thus causing $\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)$ to transition into a monotonically decreasing regime. These findings underscore the critical importance of selecting τ to balance the impact of scheduling delay and consensus failure probability. Notably, as our analysis assumes that $\tau \geq T_c$, the achievable optimum of $\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)$ may deviate from the theoretical optimum. Specifically, if $\tau^\star \geq T_c$, then $\tau^\star$ is feasible and the optimum is attained at $\tau=\tau^\star$. Otherwise, the optimum is attained at the boundary $\tau = T_c$. Intuitively, when $T_p \ll \tau$, $\tilde{\mu}_i^{\text{fail},\infty}(\tau;T_c)$ becomes monotonically decreasing with τ as the probability of consensus failure can be neglected.

B. Impact of Blockchain Consensus Process

To better understand how T_c and T_p influence the selection of scheduling interval τ, we derive the following theorem.

Proposition 1. *The optimal scheduling interval $\tau^\star$ exhibits the following strict monotonicity: it decreases with the channel coherence time T_c and increases with the blockchain propagation delay T_p.*

Proof. Please see Appendix E. ■

Theorem 1 qualitatively reveals how the two critical parameters T_c and T_p influence the optimal scheduling interval $\tau^\star$. The monotonic relationship with T_p implies that as T_p decreases, $\tau^\star$ also decreases. This occurs because shorter propagation delays reduce the probability of consensus failure, which diminishes the dominant role of T_p in the trade-off. Consequently, scheduling delay becomes the primary limiting factor, causing the average throughput to exhibit a monotonically decreasing trend over a wider range of $\tau > 0$. On the other hand, the inverse relationship with T_c indicates that a shorter channel coherence time necessitates a larger $\tau^\star$. This arises because reduced T_c amplifies the proportion of scheduling decisions that rely on the outdated channel state information. Since the throughput contribution from these delayed scheduling decisions is relatively small, the influence of T_c on $\tau^\star$ becomes weaker. In this regime, the impact of T_p becomes more pronounced. These counteracting effects highlight the delicate balance required in parameter selection. This theorem provides a principled framework for configuring $\tau^\star$ in practical BC-PFS systems, balancing the scheduling delay-failure trade-off.

VII. Pooling Effect with Delay

In this section, we systematically evaluate the pooling effect introduced by BC-PFS in O-RAN through network utility analysis. Furthermore, we derive the conditions for achieving positive pooling gain by balancing the benefits of blockchain-enhanced collaboration against the impact of the consensus process.

For the BC-PFS, multi-operator are jointly scheduled over the shared resources, serving a total of KN users. We define the network utility of BC-PFS in multi-operator O-RAN as:

$$\tilde{U}_{\text{coop}} = \sum_{i=1}^{KN} \ln\left(\tilde{\mu}_i^{\text{fail}}\left(\tau; T_c\right)\right). \tag{29}$$

In the non-cooperative PFS scenario where operators schedule independently, there are K isolated networks, each serving N users. The operators schedule independently with real-time channel tracking, i.e., $\tau = T_c$, and the scheduling process is unaffected by consensus failure. Based on (20), the average user throughput in these isolated networks is given by $\frac{1}{N}\left(\bar{r}_i + \sigma_i g(N)\right)$. However, the effective spectrum availability per user is reduced by a factor of K due to time-division frequency sharing across networks, thus the throughput for each user scales as $\frac{1}{K}$. Therefore, the network utility of non-cooperative PFS is given by:

$$\tilde{U}_{\text{iso}} = \sum_{i=1}^{KN} \ln\left(\frac{1}{KN}\left(\bar{r}_i + \sigma_i g(N)\right)\right). \tag{30}$$

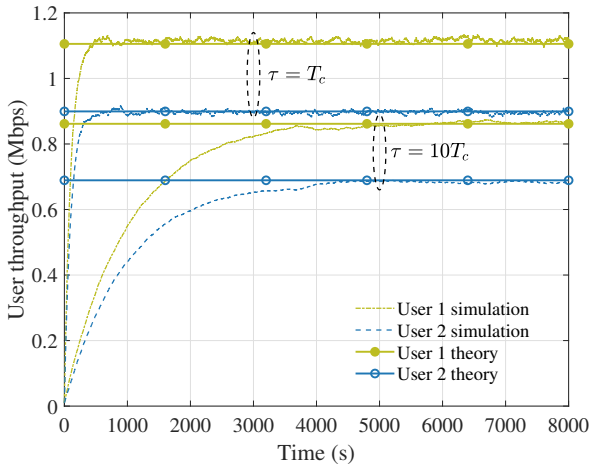


Figure 3. Simulation and theoretical results of user throughput with scheduling delay.

Building on the network utility analyzed in (29) and (30), we define the pooling gain ψ as the network utility gain of BC-PFS over non-cooperative PFS, which is given in

$$\Delta\tilde{U} \triangleq \tilde{U}_{\text{coop}} - \tilde{U}_{\text{iso}} = \sum_{i=1}^{KN} \ln\left(\frac{KN\tilde{\mu}_i^{\text{fail}}\left(\tau; T_c\right)}{\bar{r}_i + \sigma_i g\left(N\right)}\right). \tag{31}$$

While the BC-PFS in O-RAN can enhance network aggregate and achieve pooling gain, scheduling delays and consensus failure can diminish these gains. The following theorem gives the conditions under which the pooling gain can be positive.

Theorem 4. *The pooling gain is positive only if*

$$\frac{\frac{T_c}{\tau} KN g(KN,0) + \left(1-\frac{T_c}{\tau}\right)\frac{1}{\sqrt{2\pi}}}{g(N,\infty)} > 1 - exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right). \tag{32}$$

Moreover, a sufficient condition for positive pooling gain is

$$\frac{\eta_{\max} + \frac{T_c}{\tau} g(KN, \eta_{\max})}{\eta_{\max} + g(N, \eta_{\min})} > 1 - \exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right), \tag{33}$$

where $\eta_{\max} = \max_i \frac{\bar{r}_i}{\sigma_i}$ *and* $\eta_{\min} = \min_i \frac{\bar{r}_i}{\sigma_i}$.

Proof. Please see Appendix F. ■

Theorem 4 establishes the feasibility conditions for positive pooling gain introduced by BC-PFS in O-RAN. The necessary condition characterizes a requirement determined by the network scale and scheduling parameters, indicating that the benefit of cross-operator resource pooling must compensate for the performance loss caused by scheduling delay. The sufficient condition provides a conservative guarantee for a positive pooling gain by considering the extreme user statistics. Specifically, the non-cooperative PFS is evaluated using the favorable extreme $g(N, \eta_{\min})$, whereas BC-PFS is evaluated using the unfavorable extreme $g(KN, \eta_{\max})$. This constructs a pessimistic comparison for BC-PFS. If the condition is satisfied even under this conservative combination, a positive pooling gain can be guaranteed. It also highlights the need to balance the pooling advantage against the losses caused by scheduling delay and consensus failure.

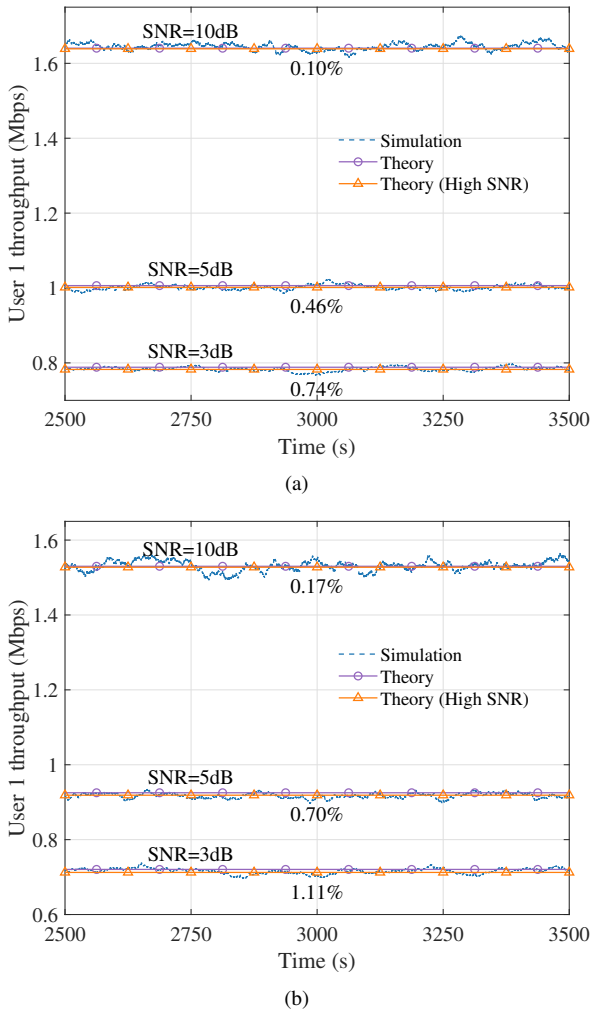


Figure 4. Simulation, theoretical results, and theoretical approximation of user throughput. (a) $\tau = 2T_c$. (b) $\tau = 5T_c$.

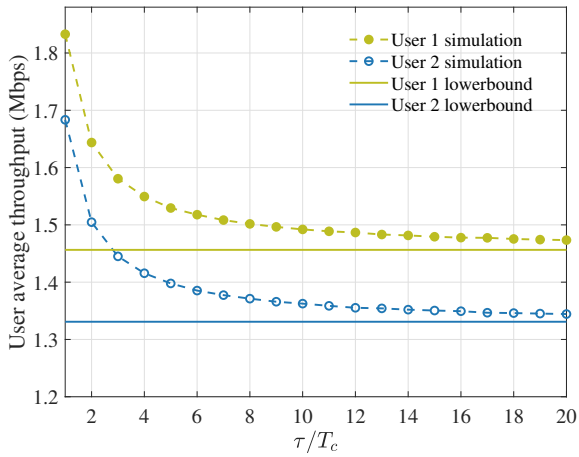


Figure 5. Simulation and theoretical lower bound of average throughput.

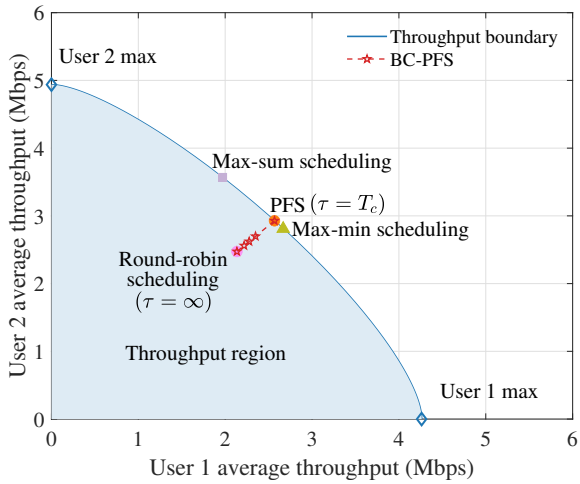


Figure 6. Throughput region: case of two users.

VIII. Simulation

In this section, we present our simulation scenarios. We adopt stationary Rayleigh fading with constant and white external noise. The smoothing factor of throughput update is set to $\alpha = 10^{-4}$, which is small enough for the BC-PFS and acceptable measure of throughput. For all scenarios, we assume the bandwidth of 1 MHz.

To validate the analysis of BC-PFS with scheduling delay presented in Section IV-C, we simulate a network with two users, each having average SNR values of 4.8 dB and 3 dB, respectively. We assume the channel coherence time $T_c = 10$ ms and examine two scheduling intervals, $\tau = T_c$ and $\tau = 10T_c$, to represent different levels of scheduling delay. Fig. 3 presents the simulation outcomes and analytic results of Theorem 2 for both users. It can be observed that for a given α, the simulated throughput update consistently demonstrates stable convergence behavior across varying conditions. The convergence values of the simulation demonstrate a strong alignment with the asymptotic analytic results. Additionally, as $\frac{\tau}{T_c}$ increases, the negative impact of delayed scheduling information becomes more evident, which reduces user throughput.

Fig. 4 presents the throughput performance of a user under different SNR conditions in a network with a total of 2 users. It compares the high-SNR theoretical results from Corollary 1, alongside the standard theoretical values from Theorem 2 and corresponding simulation results. The percentage values annotated in the figures represent the relative errors between the high-SNR theoretical values and their corresponding standard theoretical values under each SNR condition. The results demonstrate that the high-SNR theoretical approximation has broad practical applicability. Although this approximation should strictly hold only under the condition when SNR tends to infinity, simulation results reveal that even at moderate SNR levels of 3 dB, the relative error remains below 1.2%. As the SNR increases, the relative error decreases further. This phenomenon indicates that the high-SNR approximation possesses significant practical value in system design, and its applicability can reasonably be extended to moderate SNR regimes. Moreover, while increased scheduling delay does amplify the discrepancy between theoretical values, its impact remains relatively limited. Comparing the cases of $\tau = 2T_c$ in Fig. 4(a) and $\tau = 5T_c$ in Fig. 4(b), it can be observed that under the same SNR conditions, the increase in delay contributes to an error growth of no more than 0.4%.

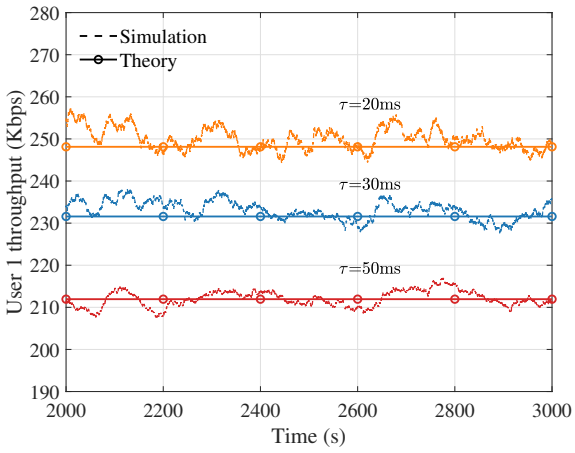


Figure 7. Simulation and theoretical results of user throughput with scheduling delay and consensus failure.

Furthermore, to validate the results of Corollary 2, Fig. 5 illustrates the relationship between the user average throughput and the ratio of scheduling interval to channel coherence time $\frac{\tau}{T_c}$. The average throughput corresponding to each ratio is calculated after the algorithm has converged. As shown in the figure, for smaller values of $\frac{\tau}{T_c}$, both users exhibit high initial throughput, which progressively decreases as $\frac{\tau}{T_c}$ increases. The throughput gradually converges toward the respective lower bounds as $\frac{\tau}{T_c}$ becomes larger. This trend highlights the increasing impact of delayed scheduling, where larger values of $\frac{\tau}{T_c}$ reduce user throughput performance. The figure confirms that the simulation results align closely with the theoretical results in Corollary 2, validating the analytic lower bounds as $\frac{\tau}{T_c} \to \infty$.

Fig. 6 presents the throughput region of a network with two users and the corresponding individual throughput achieved under different scheduling strategies. The dynamic scheduling is performed in each time slot, with only one user scheduled per slot. The figure highlights two extreme points that represent the average throughput each user would achieve if continuously scheduled over the entire duration, reflecting the maximum achievable throughput for each user. Given the Rayleigh fading channel settings, the maximum throughput corresponds to the mean user rate as defined in (1).

The maximum-sum scheduling, PFS, and max-min scheduling all operate on the throughput boundary, making these strategies Pareto optimal. That is, for any user, it is impossible to increase the throughput of one user without reducing the throughput of others [36]. Max-sum scheduling seeks to maximize the total network throughput by dynamically allocating resources based on each user's instantaneous rate, which tends to favor users with better channel conditions. In contrast, max-min scheduling prioritizes ensuring the user with the poorest channel conditions receives the most favorable scheduling, thereby maximizing the minimum throughput across all users. The PFS achieves an effective balance between fairness and overall system throughput. When $\tau = T_c$, the BC-PFS is equivalent to PFS. However, as τ increases, the individual throughput under the BC-PFS continuously decreases. When

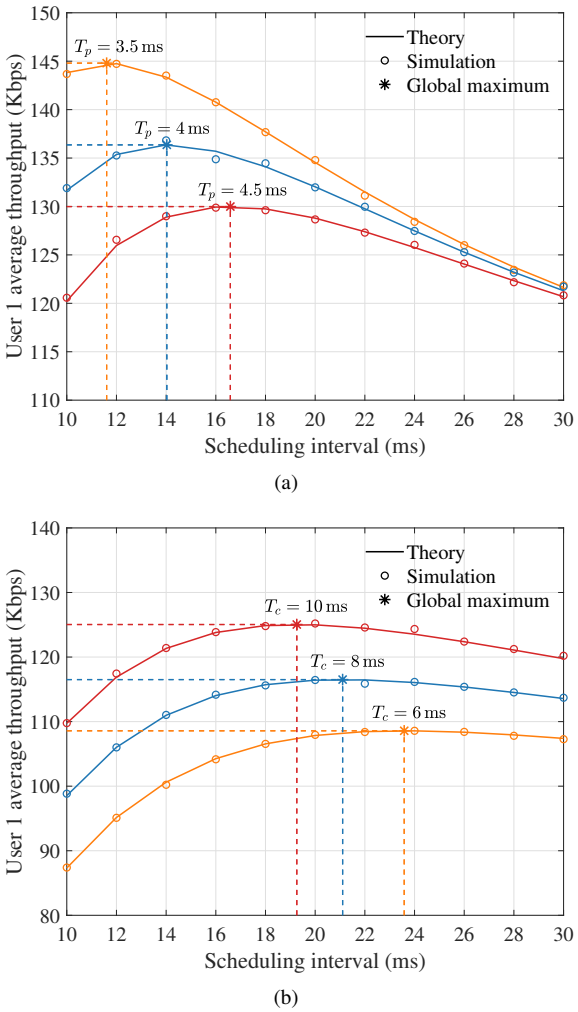


Figure 8. Illustration of the optimal scheduling interval. (a) Under different T_p. (b) Under different T_c.

$\tau \to \infty$, the throughput collapses to the level of round-robin scheduling, which allocates resources equally among users regardless of their channel states.

Furthermore, we present simulation results to validate the theoretical analysis of BC-PFS with both scheduling delay and consensus failure in Section V. Fig. 7 illustrates the user throughput performance in a network of ten users, with the channel coherence time $T_c = 10$ ms and the blockchain propagation delay $T_p = 4$ ms. The simulated throughput closely converges to the theoretical results across all scheduling intervals, which confirms the robustness of the analytical framework.

This subsection evaluates the optimal scheduling interval concerning the impact of both scheduling delay and consensus failure. Fig. 8 illustrates the average throughput performance of the network with $K = 2$, $N = 10$, where the numerically determined optimal scheduling interval for maximum throughput is explicitly identified. Fig. 8(a) examines the average throughput of one user under $T_c = 10$ ms for $T_p = 3.5$, 4, and 4.5 ms. The results show that the average throughput first increases and then decreases with longer scheduling intervals, reflecting the trade-off between wireless channel variations and

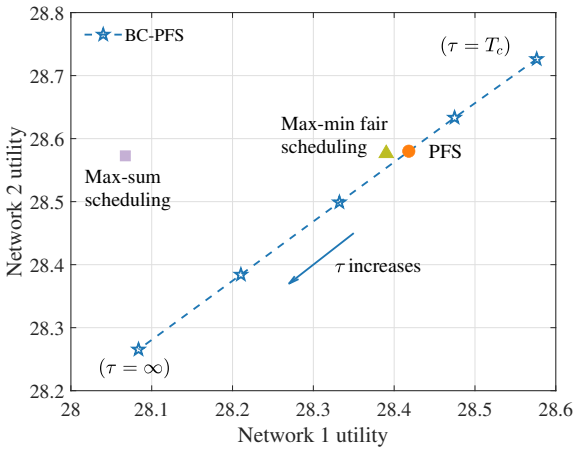


Figure 9. Utility region: case of two networks.

blockchain propagation delay, which aligns with the theoretical analysis in Theorem 3. Moreover, as T_p increases, the optimal scheduling interval that maximizes throughput shifts toward larger values, consistent with Theorem 1. Concurrently, the average throughput declines due to the higher consensus failure probability induced by longer T_p, which exacerbates throughput loss.

Fig. 8(b) investigates the case where $T_p = 5$ ms under varying $T_c = 6$, 8, and 10 ms. A similar trend of initial increase followed by a decrease in throughput is observed. Furthermore, larger T_c reduces the optimal scheduling interval for peak throughput, again in agreement with Theorem 1. The average throughput improves simultaneously because a longer T_c increases the proportion of synchronized scheduling with accurate channel information. These findings provide valuable insights for parameter optimization in practical BC-PFS systems.

This subsection examines the pooling effect through the comparison of network utility under different scheduling methods and network settings. Fig. 9 presents the utility region of two operator networks, each serving two users with $T_c = 10$ ms and $T_p = 2$ ms. The results confirm that, when operators schedule independently within their respective networks, non-cooperative PFS yields superior utility for individual networks compared to max-sum and max-min fair strategies. This is due to the inherent proportional fairness properties of PFS that maximize logarithmic utility functions. Moreover, when the two networks are aggregated under the BC-PFS, the unified system achieves substantially higher overall utility, provided the scheduling delay remains within appropriate bounds. This demonstrates that, by pooling resources and coordinating scheduling across multiple operators, the BC-PFS can enhance network performance beyond the capabilities of independent scheduling within each isolated network.

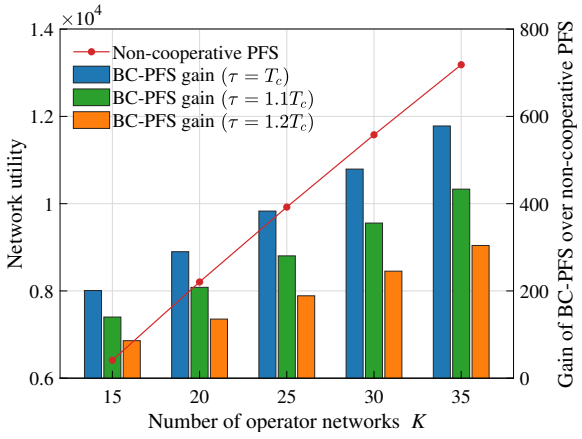
Moreover, Fig. 10 presents a comparison of the network utility between the conventional non-cooperative PFS and the BC-PFS. The line represents the network utility of non-cooperative PFS, while the bars show the utility gain of BC-PFS over non-cooperative PFS under different scheduling intervals. Fig. 10(a) is evaluated under varying numbers of operator networks K with each maintaining sixty users. The BC-PFS corresponds to the network utility of the aggregated network, while non-cooperative PFS represents the sum of the utilities of multiple independent networks. The results indicate that for $\tau = T_c$, $\tau = 1.1T_c$, and $\tau = 1.2T_c$, the network utility under the BC-PFS is significantly higher than that under non-cooperative PFS. Moreover, as the number of operator networks grows, the advantage of the BC-PFS becomes increasingly pronounced. This demonstrates that the aggregation of multiple operator networks enhances the overall network utility, particularly as the network scale expands.

Fig. 10(b) maintains twenty operator networks with varying user density N. It is obvious that the BC-PFS consistently outperforms non-cooperative PFS across the case of acceptable scheduling delays, which further validates the benefits of cooperative resource pooling. Consistent with the trend observed in Fig. 10(a), the network utility degrades as $\frac{\tau}{T_c}$ increases, reflecting the performance penalty caused by higher scheduling delays. Meanwhile, when the user density grows, both scheduling methods exhibit improved network utility due to multi-user diversity gain, while the performance gap between them, namely the pooling gain ψ, also widens. This underscores the performance advantages of the BC-PFS in densely populated networks.

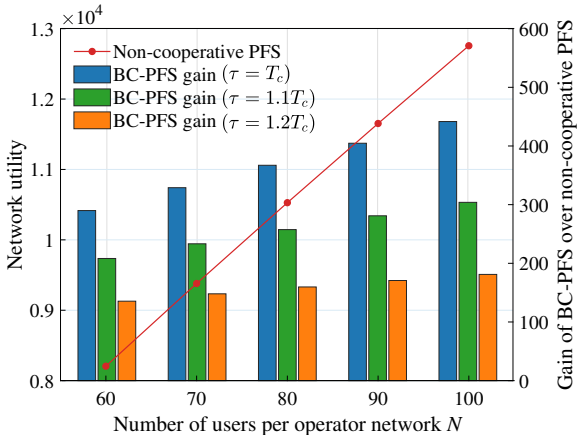
Furthermore, we evaluate the pooling effect of BC-PFS with scheduling delay and consensus failure. Fig. 11 examines the relationship between network utility and scheduling intervals under the network configuration with $T_p = 4$ ms, $T_c = 10$ ms, and $N = 2$. The results demonstrate that the BC-PFS consistently outperforms non-cooperative PFS across different operator counts and scheduling intervals, validating the fundamental advantage of resource pooling among multiple operators. Notably, the network utility of the BC-PFS follows a similar non-monotonic pattern with respect to τ as the average throughput. As τ increases, the network utility first increases and then declines. The optimal performance occurs when the trade-off between wireless channel variations and blockchain propagation delays is balanced. These insights verify the theoretical analysis in Section VII and highlight the importance of carefully selecting scheduling intervals to enhance the performance of blockchain-enhanced O-RAN.

IX. Conclusion

In this paper, we analyze the impact of blockchain consensus process on the BC-PFS for O-RAN. We first develop an analytical model for BC-PFS considering scheduling delay and prove the weak convergence of the user throughput update process. Based on the weak convergence property, we derive the semi-closed form expressions of the long-term average throughput and its approximation under high SNR or large scheduling interval. Consensus failures are further incorporated into the BC-PFS analysis framework. We uncover the trade-off between the scheduling delay and consensus failure in the BC-PFS and examine the optimal scheduling interval that balances these two factors. Furthermore, we quantify the pooling effect and provide insights for the system parameter optimization. Simulation results support our analysis and illustrate the impact of various parameters on average throughput



(a)



(b)

Figure 10. Pooling gain of BC-PFS over non-cooperative PFS. (a) Under different K. (b) Under different N.

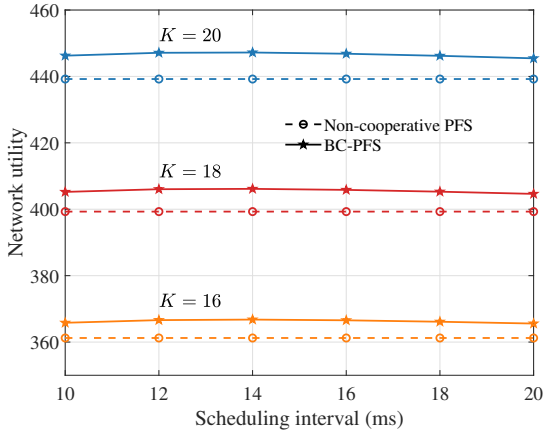


Figure 11. Comparison of network utility between non-cooperative PFS and BC-PFS under varying τ.

and network utility. This work provides crucial insights for system parameter configuration in practical implementations.

APPENDIX

A. Proof of Theorem 2

Proof. Taking steady-state expectations on both sides of (6) yields:

$$\bar{\mu}_i = \frac{T_c}{\tau} \lim_{n\to\infty} E\left[r_i\left(t_n\right) I_i\left(t_n\right)\right] + \frac{1}{\tau} \lim_{n\to\infty} E\left[\int_{T_c}^{\tau} r_i\left(t_n + u\right) du I_i\left(t_n\right)\right]. \tag{34}$$

We define c_i and d_i as scheduling metrics reflecting current and outdated channel states, respectively, given by

$$c_i \triangleq \lim_{n\to\infty} E\left[r_i\left(t_n\right) I_i\left(t_n\right)\right], \tag{35}$$

$$d_i \triangleq \lim_{n\to\infty} E\left[\int_{T_c}^{\tau} r_i\left(t_n + u\right) du I_i\left(t_n\right)\right]. \tag{36}$$

We will first analyze the component c_i. Based on the Bayesian decomposition and scheduling rule in (3), we derive:

$$\begin{aligned} c_i &= \lim_{n\to\infty} E\left[r_i\left(t_n\right) | I_i\left(t_n\right) = 1\right] \Pr\left(I_i\left(t_n\right) = 1\right) \\ &= \lim_{n\to\infty} \int_0^{\infty} x f_{r_i}\left(x\right) \Pr\left(\frac{x}{\mu_i\left(t_n\right)} > \frac{r_j\left(t_n\right)}{\mu_j\left(t_n\right)}, \forall j \neq i\right) dx, \end{aligned} \tag{37}$$

where $f_r\left(x\right)$ represents the probability density function (PDF) of user rate. From Theorem 1, $\mu_i\left(t_n\right)$ converges to $\bar{\mu}_i$ as $t \to \infty$. Therefore, taking the limit, we obtain:

$$\begin{aligned} c_i &= \int_0^{\infty} x f_{r_i}\left(x\right) \Pr\left(\frac{r_j \bar{\mu}_i}{\bar{\mu}_j} < x, \forall j \neq i\right) dx \\ &= \int_0^{\infty} x f_{r_i}\left(x\right) \prod_{\forall j \neq i} F_{r_j}\left(\frac{\bar{\mu}_j}{\bar{\mu}_i} x\right) dx, \end{aligned} \tag{38}$$

where $F_r\left(x\right)$ represents the cumulative distribution function (CDF) of the user rate.

Meanwhile, as demonstrated in [32], given $\bar{r}_i \leq \bar{r}_j$ for two users i, j, we have $\frac{\sigma_j}{\sigma_i} \leq \frac{\bar{\mu}_j}{\bar{\mu}_i} \leq \frac{\bar{r}_j}{\bar{r}_i}$ under the proportional fairness definition, where $\bar{\mu}_i$ and $\bar{\mu}_j$ are the average throughput of users i, j, respectively. Therefore, since $F_r\left(x\right)$ is monotonically non-decreasing, we can have the expressions through the variable substitution $y = \frac{x - \bar{r}_i}{\sigma_i}$:

$$\begin{aligned} c_i \leq \sigma_i &\int_{-\frac{\bar{r}_i}{\sigma_i}}^{\infty} \left(y\sigma_i + \bar{r}_i\right) f_{r_i}\left(y\sigma_i + \bar{r}_i\right) \\ &\cdot \prod_{\forall \bar{r}_i \leq \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\bar{r}_j}{\bar{r}_i} x\right) \prod_{\forall \bar{r}_i > \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\sigma_j}{\sigma_i} x\right) dy, \end{aligned} \tag{39}$$

$$\begin{aligned} c_i \geq \sigma_i &\int_{-\frac{\bar{r}_i}{\sigma_i}}^{\infty} \left(y\sigma_i + \bar{r}_i\right) f_{r_i}\left(y\sigma_i + \bar{r}_i\right) \\ &\cdot \prod_{\forall \bar{r}_i \leq \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\sigma_j}{\sigma_i} x\right) \prod_{\forall \bar{r}_i > \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\bar{r}_j}{\bar{r}_i} x\right) dy. \end{aligned} \tag{40}$$

Therefore, c_i can be approximated by the following expression that lies between above bounds:

$$c_i \approx \sigma_i \int_{-\frac{\bar{r}_i}{\sigma_i}}^{\infty} \left(y\sigma_i + \bar{r}_i\right) f_{r_i}\left(y\sigma_i + \bar{r}_i\right) \prod_{\forall j \neq i} F_{r_j}\left(y\sigma_i + \bar{r}_i\right) dy. \tag{41}$$

In fact, the user rate over Rayleigh fading channels can be accurately modeled by a normal distribution [32], i.e. approximately $r_i \sim \mathcal{N}\left(\bar{r}_i, \sigma_i^2\right)$. For normally distributed r_i, its PDF and CDF can be written as:

$$f_{r_i}(x) = \frac{1}{\sigma_i}\phi\left(\frac{x-\bar{r}_i}{\sigma_i}\right),\ F_{r_i}(x) = \Phi\left(\frac{x-\bar{r}_i}{\sigma_i}\right), \tag{42}$$

where $\bar{r}_i$ and σ_i^2 are given by (1) and (2), respectively. Substituting (42) into (41) yields:

$$c_i \approx \sigma_i \int_{-\frac{\bar{r}_i}{\sigma_i}}^{\infty} y\phi(y)\,\Phi(y)^{KN-1}\, dy + \frac{\bar{r}_i}{KN}\left(1 - \Phi\left(-\frac{\bar{r}_i}{\sigma_i}\right)^{KN}\right). \tag{43}$$

Since $\Phi\left(-\frac{\bar{r}_i}{\sigma_i}\right)^{KN} \ll 1$, (43) is approximated by

$$c_i \approx \sigma_i \int_{-\frac{\bar{r}_i}{\sigma_i}}^{\infty} y\phi(y)\,\Phi(y)^{KN-1}\, dy + \frac{\bar{r}_i}{KN} \triangleq \tilde{c}_i. \tag{44}$$

Meanwhile, the delayed component of the average throughput, namely d_i, can be written as:

$$d_i = \lim_{n\to\infty} E\left[\int_{T_c}^{\tau} r_i\left(t_n + u\right) du | I_i\left(t_n\right) = 1\right] \Pr\left(I_i\left(t_n\right) = 1\right). \tag{45}$$

With the similar analysis as c_i, we have:

$$\begin{aligned} d_i =& \left(\tau - T_c\right)\int_0^{\infty} x f_{r_i}(x)\, dx \int_0^{\infty} f_{r_i}(x) \prod_{\forall j\neq i} F_{r_j}\left(\frac{\bar{\mu}_j}{\bar{\mu}_i}x\right) dx = \\ & \frac{\tau - T_c}{KN}\left(\sigma_i\phi\left(\frac{\bar{r}_i}{\sigma_i}\right) + \bar{r}_i\Phi\left(\frac{\bar{r}_i}{\sigma_i}\right)\right)\left(1 - \Phi\left(-\frac{\bar{r}_i}{\sigma_i}\right)^{KN}\right) \approx \\ & \frac{\tau - T_c}{KN}\left(\sigma_i\phi\left(\frac{\bar{r}_i}{\sigma_i}\right) + \bar{r}_i\Phi\left(\frac{\bar{r}_i}{\sigma_i}\right)\right) \triangleq \tilde{d}_i. \end{aligned} \tag{46}$$

According to (34), the approximation of average throughput $\tilde{\mu}_i(\tau; T_c)$ can be written as:

$$\tilde{\mu}_i(\tau; T_c) = \frac{T_c}{\tau}\tilde{c}_i + \frac{1}{\tau}\tilde{d}_i. \tag{47}$$

Notably, combining (44) and (46), we can obtain that (47) is equivalent to (20), which completes the proof. ■

B. Proof of Corollary 1

Proof. $\frac{\bar{r}_i}{\sigma_i} \to \infty$ as $\gamma_i \to \infty$. Since $\lim_{\frac{\bar{r}_i}{\sigma_i}\to\infty} \Phi\left(\frac{\bar{r}_i}{\sigma_i}\right) = 1$ and $\lim_{\frac{\bar{r}_i}{\sigma_i}\to\infty} \phi\left(\frac{\bar{r}_i}{\sigma_i}\right) = 0$, (20) can be further simplified to yield the following result

$$\begin{aligned} \tilde{\mu}_i^{\infty}(\tau; T_c) &= \left(1 - \frac{T_c}{\tau}\right)\frac{\bar{r}_i}{KN} + \frac{T_c}{KN\tau} \\ &\quad \cdot \left(\bar{r}_i + \sigma_i g\left(KN, \infty\right)\right) \\ &= \frac{\bar{r}_i}{KN} + \frac{T_c}{KN\tau}\sigma_i g\left(KN, \infty\right). \end{aligned} \tag{48}$$

■

C. Proof of Corollary 2

Proof. Taking the limit on (20), we have:

$$\begin{aligned} \lim_{\tau\to\infty} \tilde{\mu}_i(\tau; T_c) &= \lim_{\tau\to\infty} \frac{T_c}{KN\tau}\left(\bar{r}_i + \sigma_i g\left(KN, \frac{\bar{r}_i}{\sigma_i}\right)\right) \\ &\quad + \lim_{\tau\to\infty} \frac{\left(1 - \frac{T_c}{\tau}\right)}{KN}\left(\bar{r}_i\Phi\left(\frac{\bar{r}_i}{\sigma_i}\right) + \sigma_i\phi\left(\frac{\bar{r}_i}{\sigma_i}\right)\right) \\ &= \frac{1}{KN}\left(\bar{r}_i\Phi\left(\frac{\bar{r}_i}{\sigma_i}\right) + \sigma_i\phi\left(\frac{\bar{r}_i}{\sigma_i}\right)\right). \end{aligned} \tag{49}$$

■

D. Proof of Theorem 3

Proof. Let $\tilde{\tau} = \frac{\tau}{T_p}$, then the first-order derivative of $\tilde{\mu}_i^{\text{fail},\infty}(\tilde{\tau}; T_c)$ with respect to $\tilde{\tau}$ is given by in (50). Denote $A = \bar{r}_i\left(\sigma_i g\left(KN, \infty\right)\right)^{-1}$, and define

$$\varphi(\tilde{\tau}) = \frac{AT_p}{T_c}\tilde{\tau}^3 + \tilde{\tau}^2 + \tilde{\tau} + 1 - \exp(\tilde{\tau}). \tag{51}$$

To analyze the stationary points of $\tilde{\mu}_i^{\text{fail},\infty}(\tilde{\tau}; T_c)$, we examine the behavior of $\varphi(\tilde{\tau})$. The first-order derivative of $\varphi(\tilde{\tau})$ is

$$\varphi'(\tilde{\tau}) = \frac{3AT_p}{T_c}\tilde{\tau}^2 + 2\tilde{\tau} + 1 - \exp(\tilde{\tau}). \tag{52}$$

The second-order derivative can then be obtained:

$$\varphi''(\tilde{\tau}) = \frac{6AT_p}{T_c}\tilde{\tau} + 2 - \exp(\tilde{\tau}). \tag{53}$$

Since $\lim_{\tilde{\tau}\to 0^+}\varphi''(\tilde{\tau}) = 1$ and $\lim_{\tilde{\tau}\to+\infty}\varphi''(\tilde{\tau}) = -\infty$, it is obvious that $\varphi''(\tilde{\tau}) = 0$ has a unique root, denoted as $\tilde{\tau}_0^{(2)}$. Consequently, $\varphi'(\tilde{\tau})$ is monotonically increasing on $\left(0, \tilde{\tau}_0^{(2)}\right]$ and monotonically decreasing on $\left[\tilde{\tau}_0^{(2)}, +\infty\right)$.

Similarly, since $\lim_{\tilde{\tau}\to 0^+}\varphi'(\tilde{\tau}) = 0$ and $\lim_{\tilde{\tau}\to+\infty}\varphi'(\tilde{\tau}) = -\infty$, $\varphi'(\tilde{\tau}) = 0$ also has a unique root, denoted as $\tilde{\tau}_0^{(1)}$. Therefore, $\varphi(\tilde{\tau})$ is monotonically increasing on $\left(0, \tilde{\tau}_0^{(1)}\right]$ and monotonically decreasing on $\left[\tilde{\tau}_0^{(1)}, +\infty\right)$.

Furthermore, since $\lim_{\tilde{\tau}\to 0^+}\varphi(\tilde{\tau}) = 0$ and $\lim_{\tilde{\tau}\to+\infty}\varphi(\tilde{\tau}) = -\infty$, the equation $\varphi(\tilde{\tau}) = 0$ has a unique root $\tilde{\tau}_0$, which satisfies $\varphi(\tilde{\tau}) > 0$ for $\tilde{\tau} < \tilde{\tau}_0$ and $\varphi(\tilde{\tau}) < 0$ for $\tilde{\tau} > \tilde{\tau}_0$. Since $\frac{\exp(-\tilde{\tau})T_c\sigma_i}{T_p\tilde{\tau}^2 KN} g\left(KN, \infty\right) > 0$ for all $\tilde{\tau} > 0$, $\partial\tilde{\mu}_i^{\text{fail},\infty}(\tilde{\tau}; T_c)/\partial\tilde{\tau} = 0$ has a unique solution at $\tilde{\tau} = \tilde{\tau}_0$. In this case, $\tilde{\mu}_i^{\text{fail},\infty}(\tilde{\tau}; T_c)$ is monotonically

$$\frac{\partial \tilde{\mu}_i^{\infty}\left(\tilde{\tau};T_c,T_p\right)}{\partial \tilde{\tau}}=\frac{\exp\left(-\tilde{\tau}\right)T_c\sigma_i}{T_p\tilde{\tau}^2KN}g\left(KN,\infty\right)\left(1-\exp\left(\tilde{\tau}\right)+\tilde{\tau}+\tilde{\tau}^2+\frac{T_p\bar{r}_i\tilde{\tau}^3}{T_c\sigma_i g\left(KN,\infty\right)}\right). \tag{50}$$

increasing on $(0,\tilde{\tau}_0]$ and monotonically decreasing on $[\tilde{\tau}_0,+\infty)$. Hence, $\tilde{\mu}_i^{\mathrm{fail},\infty}\left(\tilde{\tau};T_c\right)$ achieves its global maximum at $\tilde{\tau}=\tilde{\tau}_0$.

Finally, reverting to τ, we conclude that $\tilde{\mu}_i^{\mathrm{fail},\infty}\left(\tau;T_c\right)$ is monotonically increasing on $(0,\tau^\star]$ and monotonically decreasing on $[\tau^\star,+\infty)$, where $\tau^\star=T_p\tilde{\tau}_0$. The global maximum of $\tilde{\mu}_i^{\infty}\left(\tau;T_c,T_p\right)$ is attained at $\tau=\tau^\star$. ■

E. Proof of Proposition 1

Proof. From the proof of Theorem 3, the stationary point $\tau^\star$ satisfies

$$\frac{A}{T_cT_p^2}\tau^{\star 3}+\frac{\tau^{\star 2}}{T_p^2}+\frac{\tau^\star}{T_p}+1-\exp\left(\frac{\tau^\star}{T_p}\right)=0. \tag{54}$$

As $\tau^\star$ implicitly depends on both T_p and T_c, we may compute the partial derivatives with respect to these parameters to analyze this dependence. The partial derivatives yield:

$$\frac{\partial\tau^\star}{\partial T_p}=\frac{\tau^\star}{T_p}\cdot\frac{2A\tau^{\star 2}+2T_c\tau^\star+T_pT_c\left(1-\exp\left(\frac{\tau^\star}{T_p}\right)\right)}{3A\tau^{\star 2}+2T_c\tau^\star+T_pT_c\left(1-\exp\left(\frac{\tau^\star}{T_p}\right)\right)}, \tag{55}$$

$$\frac{\partial\tau^\star}{\partial T_c}=\frac{A\tau^{\star 3}}{T_c\left(3A\tau^{\star 2}+2T_c\tau^\star+T_pT_c\left(1-\exp\left(\frac{\tau^\star}{T_p}\right)\right)\right)}. \tag{56}$$

Let $D\triangleq 3A\tau^{\star 2}+2T_c\tau^\star+T_pT_c\left(1-\exp\left(\frac{\tau^\star}{T_p}\right)\right)$ denote common denominator in (55) and (56). As demonstrated in the proof of Theorem 3, we have $\varphi'\left(\frac{\tau^\star}{T_p}\right)=\varphi'\left(\tilde{\tau}_0\right)<0$. Expressing $\varphi'\left(\frac{\tau^\star}{T_p}\right)$ using (52) gives:

$$\varphi'\left(\frac{\tau^\star}{T_p}\right)=\frac{3A}{T_pT_c}\tau^{\star 2}+2\frac{\tau^\star}{T_p}+1-\exp\left(\frac{\tau^\star}{T_p}\right)=\frac{D}{T_pT_c}<0, \tag{57}$$

confirming that $D<0$. Notably, the numerator of $\frac{\partial\tau^\star}{\partial T_p}$ in (55) is also negative. Consequently, $\frac{\partial\tau^\star}{\partial T_p}>0$ and $\frac{\partial\tau^\star}{\partial T_c}<0$, which proves that the optimal scheduling interval $\tau^\star$ monotonically decreases with T_c and increases with T_p. ■

F. Proof of Theorem 5

Proof. We first derive the necessary condition for positive pooling gain. A necessary condition is that the upper bound of (31) is positive. According to

$$\begin{aligned}&\bar{r}_i\Phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)+\sigma_i\phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)=\bar{r}_i+\sigma_i\phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)-\bar{r}_i\left(1-\Phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)\right)\\&\le\bar{r}_i+\sigma_i\phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)\le\bar{r}_i+\tfrac{\sigma_i}{\sqrt{2\pi}},\end{aligned} \tag{58}$$

we have

$$\begin{aligned}&\tfrac{T_c}{\tau}\left(\bar{r}_i+\sigma_i g\left(KN,\tfrac{\bar{r}_i}{\sigma_i}\right)\right)+\left(1-\tfrac{T_c}{\tau}\right)\left(\bar{r}_i\Phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)+\sigma_i\phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)\right)\\&\le\bar{r}_i+\tfrac{T_c}{\tau}\sigma_i g\left(KN,\tfrac{\bar{r}_i}{\sigma_i}\right)+\left(1-\tfrac{T_c}{\tau}\right)\tfrac{\sigma_i}{\sqrt{2\pi}}.\end{aligned} \tag{59}$$

Differentiating $g\left(N,\eta\right)$ with respect to η gives

$$\frac{dg\left(N,\eta\right)}{d\eta}=-N\eta\phi(\eta)\Phi(-\eta)^{N-1}\le 0. \tag{60}$$

Thus, $g\left(KN,\frac{\bar{r}_i}{\sigma_i}\right)\le g\left(KN,0\right)$ and $g\left(N,\frac{\bar{r}_i}{\sigma_i}\right)\ge g\left(N,\infty\right)$. The argument of logarithmic term in (31) is upper bounded by

$$1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)\frac{\frac{\bar{r}_i}{\sigma_i}+\frac{T_c}{\tau}g\left(KN,0\right)+\frac{1-\frac{T_c}{\tau}}{\sqrt{2\pi}}}{\frac{\bar{r}_i}{\sigma_i}+g\left(N,\infty\right)},$$

which must exceed one for positive pooling gain to be feasible. A necessary requirement is

$$\frac{\frac{T_c}{\tau}KNg(KN,0)+\left(1-\frac{T_c}{\tau}\right)\frac{1}{\sqrt{2\pi}}}{g(N,\infty)}>1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right). \tag{61}$$

We further derive a sufficient condition for positive pooling gain by requiring that each logarithmic term in (31) is positive, i.e.,

$$\begin{aligned}&\left(\tfrac{T_c}{\tau}\left(\bar{r}_i+\sigma_i g\left(KN,\tfrac{\bar{r}_i}{\sigma_i}\right)\right)+\left(1-\tfrac{T_c}{\tau}\right)\left(\bar{r}_i\Phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)+\sigma_i\phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)\right)\right)\\&\cdot\left(1-\exp\left(-\tfrac{\tau}{T_p}\right)\left(1+\tfrac{\tau}{T_p}\right)\right)>\bar{r}_i+\sigma_i g\left(N,\tfrac{\bar{r}_i}{\sigma_i}\right).\end{aligned} \tag{62}$$

According to

$$\sigma_i\phi\left(\frac{\bar{r}_i}{\sigma_i}\right)-\bar{r}_i\left(1-\Phi\left(\frac{\bar{r}_i}{\sigma_i}\right)\right)=\sigma_i\int_{\frac{\bar{r}_i}{\sigma_i}}^{\infty}\left(x-\frac{\bar{r}_i}{\sigma_i}\right)\phi(x)dx>0,$$

we have $\bar{r}_i\Phi\left(\frac{\bar{r}_i}{\sigma_i}\right)+\sigma_i\phi\left(\frac{\bar{r}_i}{\sigma_i}\right)>\bar{r}_i$. Hence,

$$\begin{aligned}&\tfrac{T_c}{\tau}\left(\bar{r}_i+\sigma_i g\left(KN,\tfrac{\bar{r}_i}{\sigma_i}\right)\right)+\left(1-\tfrac{T_c}{\tau}\right)\left(\bar{r}_i\Phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)+\sigma_i\phi\left(\tfrac{\bar{r}_i}{\sigma_i}\right)\right)\\&>\bar{r}_i+\tfrac{T_c}{\tau}\sigma_i g\left(KN,\tfrac{\bar{r}_i}{\sigma_i}\right).\end{aligned} \tag{63}$$

Based on (62), a stronger sufficient condition is

$$\begin{aligned}&\left(1-\exp\left(-\tfrac{\tau}{T_p}\right)\left(1+\tfrac{\tau}{T_p}\right)\right)\left(\bar{r}_i+\tfrac{T_c}{\tau}\sigma_i g\left(KN,\tfrac{\bar{r}_i}{\sigma_i}\right)\right)\\&>\bar{r}_i+\sigma_i g\left(N,\tfrac{\bar{r}_i}{\sigma_i}\right).\end{aligned} \tag{64}$$

Equivalently,

$$1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)>\frac{\frac{\bar{r}_i}{\sigma_i}+g\left(N,\frac{\bar{r}_i}{\sigma_i}\right)}{\frac{\bar{r}_i}{\sigma_i}+\frac{T_c}{\tau}g\left(KN,\frac{\bar{r}_i}{\sigma_i}\right)} \tag{65}$$

According to (60), $\frac{dg(N,\eta)}{d\eta}\le 0$. Therefore, we have $g(N,\eta_i)\le g(N,\eta_{\min})$ and $g(KN,\eta_i)\ge g(KN,\eta_{\max})$, where $\eta_i=\frac{\bar{r}_i}{\sigma_i}$, $\eta_{\min}=\min\limits_i\frac{\bar{r}_i}{\sigma_i}$ and $\eta_{\max}=\max\limits_i\frac{\bar{r}_i}{\sigma_i}$. By considering the extreme values of heterogeneous user statistics, a conservative upper bound for the right-hand side of (65) is

$$\frac{\eta_i+g(N,\eta_{\min})}{\eta_i+\frac{T_c}{\tau}g(KN,\eta_{\max})}. \tag{66}$$

Since $1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)<1$, this bound must also be smaller than 1 for the sufficient condition to hold. This requires $\frac{T_c}{\tau}g(KN,\eta_{\max})>g(N,\eta_{\min})$. Next, we determine the maximum of this bound over η_i. Its derivative with respect to η_i is

$$\frac{\frac{T_c}{\tau}g(KN,\eta_{\max})-g(N,\eta_{\min})}{\left(\eta_i+\frac{T_c}{\tau}g(KN,\eta_{\max})\right)^2}>0.$$

Thus, we have a sufficient condition for positive pooling gain:

$$\frac{\eta_{\max}+\frac{T_c}{\tau}g(KN,\eta_{\max})}{\eta_{\max}+g(N,\eta_{\min})}>1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right). \tag{67}$$

Combining (61) and (67) completes the proof. ■

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